SRI RAMAKRISHNA INSTITUTE OF TECHNOLOGY
COIMBATORE-10
(An Autonomous Institution affiliated to Anna University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CURRICULUM STRUCTURE

B.E. – ELECTRONICS AND COMMUNICATION ENGINEERING

SEMESTER – I

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20HSG01	Technical English	HS	2	0	2	4	3
2.	20MHG01	Calculus and Linear Algebra	BS	3	1	0	4	4
3.	20PHG01	Engineering Physics	BS	3	1	0	4	4
4.	20CHG01	Engineering Chemistry	BS	3	1	0	4	4
5.	20ITG01	Programming for Problem Solving using C	ES	3	0	0	3	3
PRACTICALS								
6.	20MEG02	Engineering Workshop	ES	0	0	4	4	2
7.	20PHG02	Engineering Physics Laboratory	BS	0	0	3	3	1.5
8.	20ITG02	Programming in C Laboratory	ES	0	0	4	4	2
			Total	**14**	**3**	**13**	**30**	**23.5**

SEMESTER – II

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20HSG02	Universal Human Values II - Understanding Harmony	HS	3	0	0	3	3
2.	20MHG02	Differential Equations and Complex Variables	BS	3	1	0	4	4
3.	20CSG01	Object Oriented Programming using C++	ES	3	0	0	3	3
4.	20ECG01	Electric Circuits	ES	3	0	0	3	3
5.	20EC001	Electron Devices	PC	3	0	0	3	3
PRACTICALS								
6.	20MEG01	Engineering Graphics	ES	0	0	4	4	2
7.	20CHG02	Engineering Chemistry Laboratory	BS	0	0	3	3	1.5
8.	20CSG02	Programming in C++ Laboratory	ES	0	0	4	4	2
			Total	**18**	**1**	**11**	**30**	**21.5**
9.	20AC001	Environmental Science and Engineering	AC	3	0	0	3	-

SEMESTER – III

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20MHG03	Transforms and Partial Differential Equations	BS	3	1	0	4	4

2.	20EEG05	Electrical Engineering	ES	2	0	2	4	3
3.	20EC002	Analog Electronics	PC	3	0	0	3	3
4.	20EC003	Digital Electronics	PC	3	0	0	3	3
5.	20EC004	Electromagnetic Fields and Waves	PC	3	0	0	3	3
6.	20xxExx	Open Elective –I	OE	3	0	0	3	3
PRACTICALS								
7.	20EC005	Analog Electronics Laboratory	PC	0	0	3	3	1.5
8.	20EC006	Digital Electronics Laboratory	PC	0	0	3	3	1.5
			Total	**17**	**1**	**8**	**26**	**22**

SEMESTER – IV

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20MHG05	Probability and Random Processes	BS	3	0	0	3	3
2.	20EC007	Linear Integrated Circuits	PC	3	0	0	3	3
3.	20EC008	Signals and Systems	PC	3	0	0	3	3
4.	20EC009	Transmission Lines and Waveguides	PC	3	0	0	3	3
5.	20EC010	Microprocessor and Microcontroller	PC	3	0	0	3	3
6.	20xxExx	Open Elective –II	OE	3	0	0	3	3
PRACTICALS								
7.	20EC011	Integrated Circuits Laboratory	PC	0	0	3	3	1.5
8.	20EC012	Microprocessor and Microcontroller Laboratory	PC	0	0	3	3	1.5
			Total	**18**	**0**	**6**	**24**	**21**

SEMESTER – V

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20HMG05	Professional Practice and Ethics	HS	2	0	0	2	2
2.	20EC013	Digital Signal Processing	PC	3	0	0	3	3
3.	20EC014	Analog Communication	PC	3	0	0	3	3
4.	20EC015	Control Systems	PC	3	0	0	3	3
5.	20ECPxx	Professional Elective-I	PE	3	0	0	3	3
6.	20xxExx	Open Elective – III	OE	3	0	0	3	3
PRACTICALS								
7.	20EC016	Digital Signal Processing Laboratory	PC	0	0	3	3	1.5
8.	20EC017	Analog Communication Laboratory	PC	0	0	3	3	1.5
			Total	**17**	**0**	**6**	**23**	**20**

SEMESTER – VI

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20EC018	Digital VLSI Design	PC	3	0	0	3	3
2.	20EC019	Digital Communication	PC	3	0	0	3	3
3.	20EC020	Antennas and Wave Propagation	PC	3	0	0	3	3
4.	20ECPxx	Professional Elective-II	PE	3	0	0	3	3
5.	20xxExx	Open Elective –IV	OE	3	0	0	3	3
PRACTICALS								
6.	20EC021	VLSI Design Laboratory	PC	0	0	3	3	1.5
7.	20EC022	Digital	PC	0	0	3	3	1.5

		Communication Laboratory						
8.	20EC901	Design Project	EC	0	0	6	6	3
			Total	**18**	**0**	**12**	**30**	**21**
9.	20AC002	Constitution of India	AC	3	0	0	3	-

SEMESTER – VII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20EC023	Optical Communication	PC	3	0	0	3	3
2.	20EC024	RF and Microwave Communication	PC	3	0	0	3	3
3.	20ECPxx	Professional Elective-III	PE	3	0	0	3	3
4.	20ECPxx	Professional Elective-IV	PE	3	0	0	3	3
5.	20ECPxx	Professional Elective-V	PE	3	0	0	3	3
PRACTICALS								
6.	20EC025	Optical and Microwave Laboratory	PC	0	0	4	4	2
7.	20EC902	Final Year Project - I	EC	0	0	6	6	3
			Total	**15**	**0**	**10**	**25**	**20**

SEMESTER – VIII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	20ECPxx	Professional Elective-VI	PE	3	0	0	3	3
PRACTICALS								
2.	20EC903	Final Year Project - II	EC	0	0	16	16	8
			Total	**3**	**0**	**16**	**19**	**11**

TOTAL NUMBER OF CREDITS: 160

PROFESSIONAL ELECTIVES

ELECTIVE – I & II GROUP

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	20ECP01	Electronic Measurements	PE	3	0	0	3	3
2.	20ECP02	Advanced Digital System Design	PE	3	0	0	3	3
3.	20ECP03	Computer Architecture	PE	3	0	0	3	3
4.	20ECP04	Advanced Microprocessors and Microcontrollers	PE	3	0	0	3	3
5.	20ECP05	Optoelectronics	PE	3	0	0	3	3
6.	20ECP06	Biomedical Instrumentation	PE	3	0	0	3	3
7.	20ECP07	Numerical Analysis	PE	3	0	0	3	3
8.	20ECP08	Nano Electronics	PE	3	0	0	3	3
9.	20ECP09	Electromagnetic Interference and Compatibility	PE	3	0	0	3	3
10.	20ECP10	Telecommunication Switching and Networks	PE	3	0	0	3	3

ELECTIVE –III & IV GROUP

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	20ECP11	Digital Image Processing	PE	3	0	0	3	3
2.	20ECP12	Principles of Satellite Communication	PE	3	0	0	3	3
3.	20ECP13	High Performance Communication Networks	PE	3	0	0	3	3

4.	20ECP14	Real time Concepts for Embedded Systems	PE	3	0	0	3	3
5.	20ECP15	Software for Embedded Systems	PE	3	0	0	3	3
6.	20ECP16	Artificial Intelligence	PE	3	0	0	3	3
7.	20ECP17	Low Power VLSI Design	PE	3	0	0	3	3
8.	20ECP18	Optical Networks	PE	3	0	0	3	3
9.	20ECP19	Electronic Packaging	PE	3	0	0	3	3
10.	20ECP20	Introduction to MEMS System Design	PE	3	0	0	3	3

ELECTIVE – V & VI GROUP

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	20ECP21	Wireless Communication	PE	3	0	0	3	3
2.	20ECP22	Cognitive Radio	PE	3	0	0	3	3
3.	20ECP23	Wireless Networks	PE	3	0	0	3	3
4.	20ECP24	Cryptography and Network Security	PE	3	0	0	3	3
5.	20ECP25	Wireless Sensor Networks	PE	3	0	0	3	3
6.	20ECP26	Introduction to Embedded Controllers	PE	3	0	0	3	3
7.	20ECP27	ASIC Design	PE	3	0	0	3	3
8.	20ECP28	Machine Learning	PE	3	0	0	3	3
9.	20ECP29	Deep Learning	PE	3	0	0	3	3
10.	20ECP30	Cognitive Robotics	PE	3	0	0	3	3

OPEN ELECTIVES

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
OPEN ELECTIVES OFFERED BY CSE DEPARTMENT TO OTHER B.E.PROGRAMMES								
1	20CSE01	Basics of Python Programming	OE	3	0	0	3	3
2	20CSE02	Introduction to AI	OE	3	0	0	3	3
3	20CSE03	Fundamentals of Data Science	OE	3	0	0	3	3
4	20CSE04	Basics of Internet Programming	OE	3	0	0	3	3
5	20CSE05	Introduction to Soft Computing	OE	3	0	0	3	3
OPEN ELECTIVES OFFERED BY EEE DEPARTMENT TO OTHER B.E.PROGRAMMES								
1	20EEE01	Energy Management Systems	OE	3	0	0	3	3
2	20EEE02	Medical Instrumentation	OE	3	0	0	3	3
3	20EEE03	PLC Programming	OE	3	0	0	3	3
4	20EEE04	Renewable Energy Systems	OE	3	0	0	3	3
5	20EEE05	Virtual Instrumentation & Data Acquisition	OE	3	0	0	3	3
6	20EEE06	Electric Vehicles	OE	3	0	0	3	3
OPEN ELECTIVES OFFERED BY IT DEPARTMENT TO OTHER B.E.PROGRAMMES								
1	20ITE01	Big Data Analytics and its Applications	OE	3	0	0	3	3
2	20ITE02	Cloud Computing Fundamentals	OE	3	0	0	3	3
3	20ITE03	Fundamentals of Internet of Things	OE	3	0	0	3	3

4	20ITE04	Introduction to Database Management Systems	OE	3	0	0	3	3
5	20ITE05	Web Interface Design and Development	OE	3	0	0	3	3
6	20ITE06	Introduction to Data Structures	OE	3	0	0	3	3
7	20ITE07	Principles of Software Engineering	OE	3	0	0	3	3
OPEN ELECTIVES OFFERED BY MECHANICAL ENGINEERING DEPARTMENT TO OTHER B.E.PROGRAMMES								
1	20MEE01	Automotive Fundamentals	OE	3	0	0	3	3
2	20MEE02	Computer Aided Design	OE	3	0	0	3	3
3	20MEE03	Power Plant Engineering	OE	3	0	0	3	3
4	20MEE04	Industrial Engineering	OE	3	0	0	3	3
5	20MEE05	Rapid Prototyping	OE	3	0	0	3	3
OPEN ELECTIVES OFFERED BY ECE DEPARTMENT TO OTHER B.E.PROGRAMMES								
1	20ECE01	Electronic Measurements and Instrumentation	OE	3	0	0	3	3
2	20ECE02	Microcontrollers and its Applications	OE	3	0	0	3	3
3	20ECE03	Introduction to Embedded Systems	OE	3	0	0	3	3
4	20ECE04	Nano Electronics and Sensors	OE	3	0	0	3	3
5	20ECE05	Principles of VLSI Systems	OE	3	0	0	3	3

CREDIT DISTRIBUTION

S. No.	Course Work - Subject Area	Credits/Semester								Credits Total
		I	II	III	IV	V	VI	VII	VIII	
1	Humanities and Social Sciences including Management Courses (HSMC)	3	3	0	0	2	0	0	0	**8**
2	Basic Sciences Courses (BS)	13.5	5.5	4	3	0	0	0	0	**26**
3	Engineering Science Courses (ES)	7	10	3	0	0	0	0	0	**20**
4	Professional Core Courses (PC)	0	3	12	15	12	12	8	0	**62**
5	Professional Elective Courses (PE)	0	0	0	0	3	3	9	3	**18**
6	Open Electives (OE)	0	0	3	3	3	3	0	0	**12**
7	Employability Enhancement Courses (EC)	0	0	0	0	0	3	3	8	**14**
	Total	**23.5**	**21.5**	**22**	**21**	**20**	**21**	**20**	**11**	**160**

SRI KRISHNA INSTITUTE OF TECHNOLOGY
WISDOM DISCIPLINE PROSPERITY
COIMBATORE - 10